

Finite-Scale Shadowing: Local 2-Adic Descent and the Two Missing Bridges in an Effective Collatz Bound

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Abstract

We separate two previously conflated mechanisms in an attempted effective almost-all Collatz bound. A finite-scale 2-adic counting argument rigorously controls forward descent on local scale windows, but iteration requires an open restart/transport estimate for the endpoint distribution. Independently, the tilted reverse model produces the geometric ratio $\gamma_{\text{rev}} = I_{\text{rev}}(1.33)/(\log_2 3 - 1.33) \approx 0.993$; attaching this reverse-growth exponent to the forward exceptional set requires a distinct reverse-chord encoding hypothesis. We formulate both missing bridges explicitly and show that the direct method of types gives a positive local rate for $Y \asymp N_0^\alpha$ whenever $1 < \alpha < 2/\log_2 3 \approx 1.26186$. Thus 0.993 is presently a reverse-geometric characteristic, not an established exceptional-set exponent.

Units and conventions

All rate functions, entropies and gains are evaluated consistently in a single unit; we adopt *bits*. Concretely, the tilt is 2^{sa} with $s < 1$, $\Lambda_n(s) = \log_2 \rho_n(s) - \log_2 \rho_n(0)$ (so $\Lambda(0) = 0$, $\Lambda'(0) = 2$), and $I(\sigma) = \sup_s [s\sigma - \Lambda(s)]$ is in bits. Equivalently, in nats (tilt e^{sa} , $s < \ln 2$, natural logs) $I_{\text{nat}}(\sigma) = I_{\text{bits}}(\sigma) \ln 2$. The reverse structural gain is $g = \log_2 3 - 1.33 \approx 0.25496$ bits/step $= (\ln 3 - 1.33 \ln 2) \approx 0.17673$ nats/step. The reverse-geometric ratio $\gamma_{\text{rev}} = I_{\text{rev}}/g$ is dimensionless and unit-independent: $0.2532/0.25498 \approx 0.993 = 0.17550/0.17673$. This unit conversion does not turn γ_{rev} into a forward exceptional-set exponent. The earlier value 0.67 arose from mixing a nats rate with a bits gain and is superseded. We normalize $I(2) = 0$ throughout.

Introduction

Tao's almost-all theorem relies on transporting local information between scales. The direct 2-adic distribution of a finite valuation word is rigorous, but it remains uniform only while the required cylinder modulus fits inside the sampled interval. For forward descent this yields a local window, not an unrestricted iteration. The missing ingredient is a restart/transport lemma controlling the push-forward distribution of surviving endpoints.

A second issue is directional. The value $1.33 < \log_2 3$ describes a reverse growing chord, whereas a forward segment that remains above N_0 satisfies an upper shift threshold

strictly larger than $\log_2 3$. Therefore the ratio $I_{\text{rev}}(1.33)/(\log_2 3 - 1.33)$ cannot be assigned to the forward exceptional set without an additional combinatorial encoding. This note records the local rigorous statement and isolates these two open bridges.

Finite-scale counting

Let S_d denote the sum of the first d Syracuse valuations. A valuation word of total weight S determines one 2-adic cylinder of relative density 2^{-S} among odd integers, while the number of such words is $\binom{S-1}{d-1}$.

Proposition 1 (Finite-scale 2-adic LDP). *Let $\mathcal{I}$ contain M odd integers and let $m \geq d$. Then*

$$\mathbb{P}_{N \in \mathcal{I}}(S_d \leq m) \leq \mathbb{P}(G_1 + \cdots + G_d \leq m) + O\left(\frac{\binom{m}{d}}{M}\right), \quad G_i \stackrel{\text{iid}}{\sim} \text{Geom}(1/2).$$

If $m = \sigma d$ and $m \leq \log_2 M + O(\log d)$, this implies

$$\mathbb{P}_{N \in \mathcal{I}}(S_d \leq \sigma d) \leq 2^{-dI_2(\sigma) + O(\log d)}.$$

Proof. Summing the cylinder masses and interval-counting errors over valuation words gives

$$\sum_{S=d}^m \binom{S-1}{d-1} 2^{-S} + O\left(M^{-1} \sum_{S=d}^m \binom{S-1}{d-1}\right).$$

The first sum is the stated geometric probability and the second word count is $\binom{m}{d}$. The asymptotic bound follows from Cramér's estimate and $I_2(\sigma) + \sigma H_2(1/\sigma) = \sigma$. $\square$

Corollary 2 (Unconditional local descent window). *Let $\lambda = \log_2 3$, fix $1 < \alpha < 2/\lambda$, and put $Y \asymp N_0^\alpha$. Choose*

$$\frac{\alpha - 1}{2 - \lambda} < t \leq \frac{1}{\lambda}, \quad d = \lfloor t \log_2 N_0 \rfloor, \quad \sigma = \lambda + \frac{\alpha - 1}{t} < 2.$$

The proportion of odd $N \asymp Y$ whose first d accelerated odd iterates all exceed N_0 is

$$\leq N_0^{-tI_2(\sigma) + o(1)}.$$

The rate is positive for every $\alpha < 2/\log_2 3 \approx 1.26186$ and degenerates at the endpoint.

Proof. If $x_k > N_0$ for $0 \leq k \leq d$, then the exact identity

$$x_d = N 3^d 2^{-S_d} \prod_{k < d} \left(1 + \frac{1}{3x_k}\right)$$

gives

$$S_d < d\lambda + \log_2(N/N_0) + O(d/N_0).$$

For $N \asymp N_0^\alpha$ this threshold is $\sigma d + o(d)$. The condition $t\lambda \leq 1$ makes the required cylinder modulus fit inside the sampled interval, and the lower bound on t ensures $\sigma < 2$. $\square$

The two open bridges

Lemma 3 (Within-word endpoint transport). *Fix an admissible valuation word of total weight S and length d . There are odd integers ρ_w, y_w such that*

$$N = \rho_w + 2^{S+1}q, \quad \text{Syr}^d(N) = y_w + 2 \cdot 3^d q.$$

The barrier conditions on all intermediate iterates restrict q to an integer interval. Hence in every odd residue class modulo 2^m the endpoint count differs from the uniform value by at most one.

Proof. Apply the affine word identity at every prefix. Exact valuation among odd starts fixes a class modulo 2^{S+1} , and all prefix maps are increasing in q . Since 3^d is invertible modulo 2^{m-1} , the endpoint progression permutes odd residue classes modulo 2^m . $\square$

Proposition 4 (Exact boundary-layer spectrum). *Let $K = 2^M$, use the odd coordinate $N = 2z + 1$, and set $\mathcal{B}_{d,M} = \{z : S_d(2z + 1) \leq M\}$. For an exact valuation word w of weight S , let $r_w \pmod{2^S}$ be its residue in the z coordinate. Then, for $h \neq 0$ and $v = v_2(h)$,*

$$\frac{\widehat{\mathbf{1}_{\mathcal{B}_{d,M}}}(h)}{K} = \sum_{S=\max\{d, M-v\}}^M 2^{-S} \sum_{\substack{w \in \mathbb{N}^d \\ |w|=S}} \exp(-2\pi i h r_w / K).$$

In particular, odd frequencies see only the top layer $|w| = M$.

Proof. The exact word condition is $3^d N + c_w \equiv 2^S \pmod{2^{S+1}}$, not merely divisibility by 2^S . Thus $z \equiv r_w \pmod{2^S}$. The Fourier transform of this cylinder vanishes unless $2^{M-S} \mid h$ and otherwise has normalized value $2^{-S} \exp(-2\pi i h r_w / K)$. Sum over the disjoint words with $S \leq M$. $\square$

Lemma 5 (Top-layer offset injectivity). *Fix $d \leq M$ and distinct words $w, w' \in \mathbb{N}^d$ of total weight M . Then*

$$r_w \not\equiv r_{w'} \pmod{2^{M-1}}.$$

In particular, $A_{\text{ant}} = 0$ on every fixed (d, M) top layer. This statement does not compare words of different lengths.

Proof. If S_j and S'_j are the partial sums, then

$$c_w = \sum_{j=0}^{d-1} 3^{d-1-j} 2^{S_j}.$$

At the smallest exponent in the symmetric difference of the two partial-sum sets, the difference $c_w - c_{w'}$ has an odd leading coefficient; hence $v_2(c_w - c_{w'}) < M$. The CRT formula $\rho_w \equiv (2^M - c_w)3^{-d} \pmod{2^{M+1}}$ then shows that equality of r_w and $r_{w'}$ modulo 2^{M-1} is impossible. $\square$

Lemma 6 (Spectral Transversality of Resonant Sets; falsified as a decay mechanism). *Let $R_{\text{end}} = \{h : \|h3^{d_{\text{end}}}/2^M\| \leq A/2^M\}$ be the set of resonant frequencies for the endpoint distribution, and $R_{\text{bad}} = \{h : \|h3^{d_{\text{bad}}}/2^M\| \leq A/2^M\}$ be the resonant set for the bad-set boundary layer. While the arithmetic fact $d_{\text{end}} \not\equiv d_{\text{bad}} \pmod{2^{M-2}}$ holds, enforcing that the central peaks do not intersect, this separation does not imply smallness of the total error Δ_b . The high-frequency noise cross-terms generate macroscopic peaks of magnitude ~ 0.43 independent of the resonance alignment, exceeding the required $2^{-B/2}$ threshold by a factor of 885.*

Remark 7 (Falsification of intra-layer anomalous cancellation). Extensive exact numerical DP tracking over the bucket constraints (up to $M \leq 10^4$) explicitly falsifies any hypothesis of anomalous intra-layer Fourier cancellation. Within any fixed, macroscopic slice of length $d \approx M/2$, the Fourier coefficients uniformly obey $\theta \approx 0.47 \approx 0.5$, mirroring an unstructured random walk. Thus, the decay in Δ_b is not driven by the endpoint distribution becoming artificially smooth, but rather by its sharp peaks systematically avoiding the structural resonances of the bad set.

Remark 8 (Finite diagnostic and why pointwise transport is too strong). At the full next-event modulus, many endpoint residue classes are empty, so pointwise Haar domination is numerically false and unnecessary. For $\alpha = 1.10$ and $B = 16, 18, 20$, exhaustive enumeration gives mass-weighted bad-event errors

$$0.003506, \quad 0.001901, \quad 0.000887,$$

while multiplication by $2^{B/2}$ gives 0.898, 0.973, 0.908. At $B = 20$ the weighted absolute Fourier majorant is 0.05268, versus a signed aggregate cancellation ratio 0.00911. Three scales identify the target mechanism but do not prove decay. Separate exhaustive scale-kernel computations give bucket-weighted discrepancies of order 0.5%–1.6%.

Remark 9 (Reverse-chord encoding: superseded). The Reverse-chord Encoding Hypothesis was previously formulated as a potential bridge to attach $\gamma_{\text{rev}} \approx 0.993$ to the forward exceptional set. Numerical experiments (E9) revealed a fundamental information-theoretic barrier: the entropy of reverse chords with mean shift $4/3$ is $h(4/3) \approx 1.0817$ bits/step, while the target space requires $\log_2 3 \approx 1.585$ bits/step. The resulting Hausdorff dimension deficit ($D_{\text{rev}} \approx 0.811 < D_{\text{fwd}} \approx 0.950$) makes 100% coverage algebraically impossible. This route is closed.

Proposition 10 (Conditional reverse-geometric ratio). *The reverse thermodynamic model assigns to a mean-1.33 growing chord the cost-per-gain ratio*

$$\gamma_{\text{rev}} = \frac{I_{\text{rev}}(1.33)}{\log_2 3 - 1.33} \approx \frac{0.2532}{0.25498} \approx 0.993.$$

This is a characteristic of reverse growth geometry. It becomes a candidate forward exceptional-set exponent only under both reverse LDP shadowing and the Reverse-chord Encoding Conjecture.

Remark 11 (Why one shadowing statement is insufficient). Forward non-descent gives an upper threshold

$$\frac{S_d}{d} < \log_2 3 + \frac{\log_2(N/N_0)}{d} + O(N_0^{-1}) \quad (\text{which is } > \log_2 3),$$

whereas the value $1.33 < \log_2 3$ belongs to reverse growth. Thus reverse LDP shadowing alone does not imply a forward exceptional-set estimate. The restart/transport and reverse-encoding problems are logically distinct.

Remark 12 (Honest status). The finite-scale counting proposition and local descent corollary are unconditional. No global power-law exceptional-set bound is proved here. The previously quoted residual densities at $N_0 = 2^{68}$ are only scale illustrations under the stronger, unproved reverse-encoding hypothesis. A density statement would still not imply the pointwise Collatz conjecture.